PYTHON DATA ANALYSIS – PROJECT REPORT

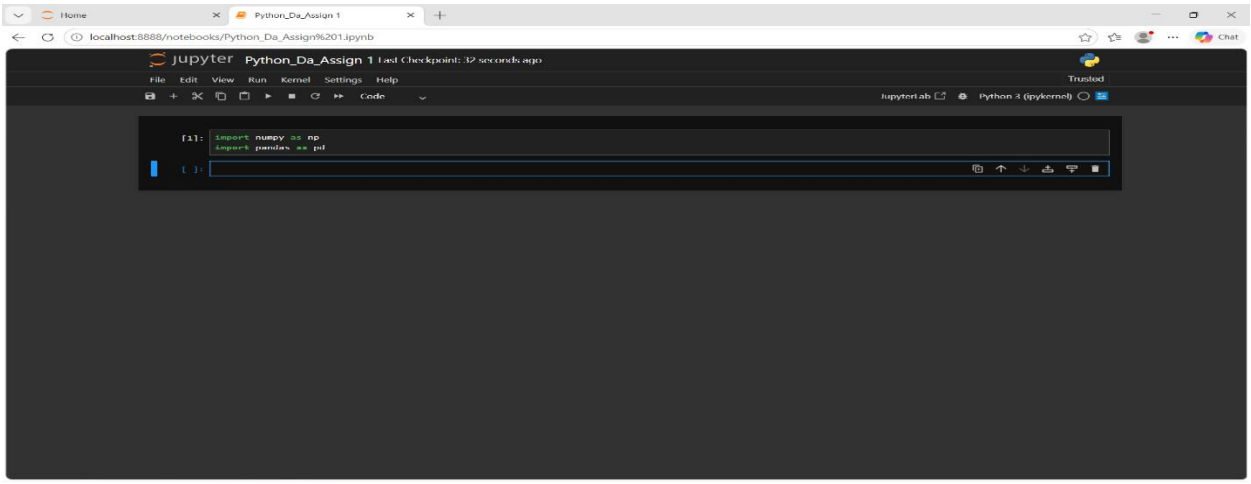
Data Analysis Using NumPy and Pandas

Submitted By: Harikarthikeyan V

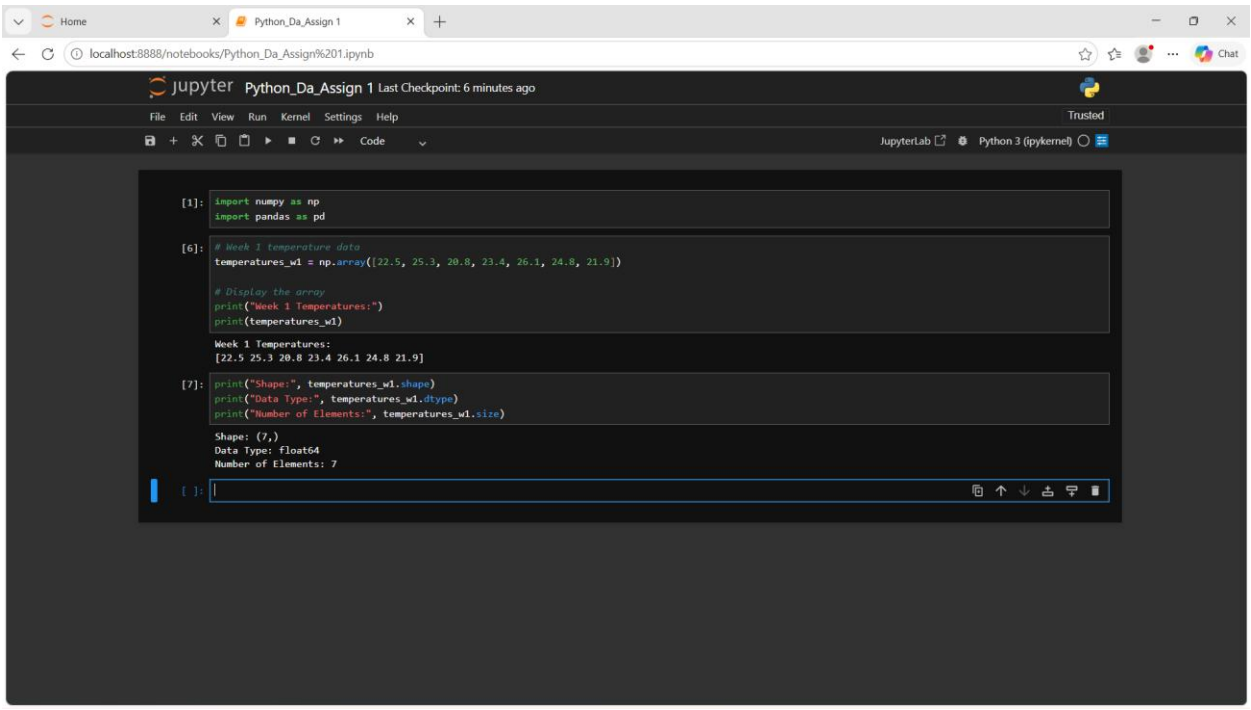
Screenshots

Task1: NumPy Array Operations

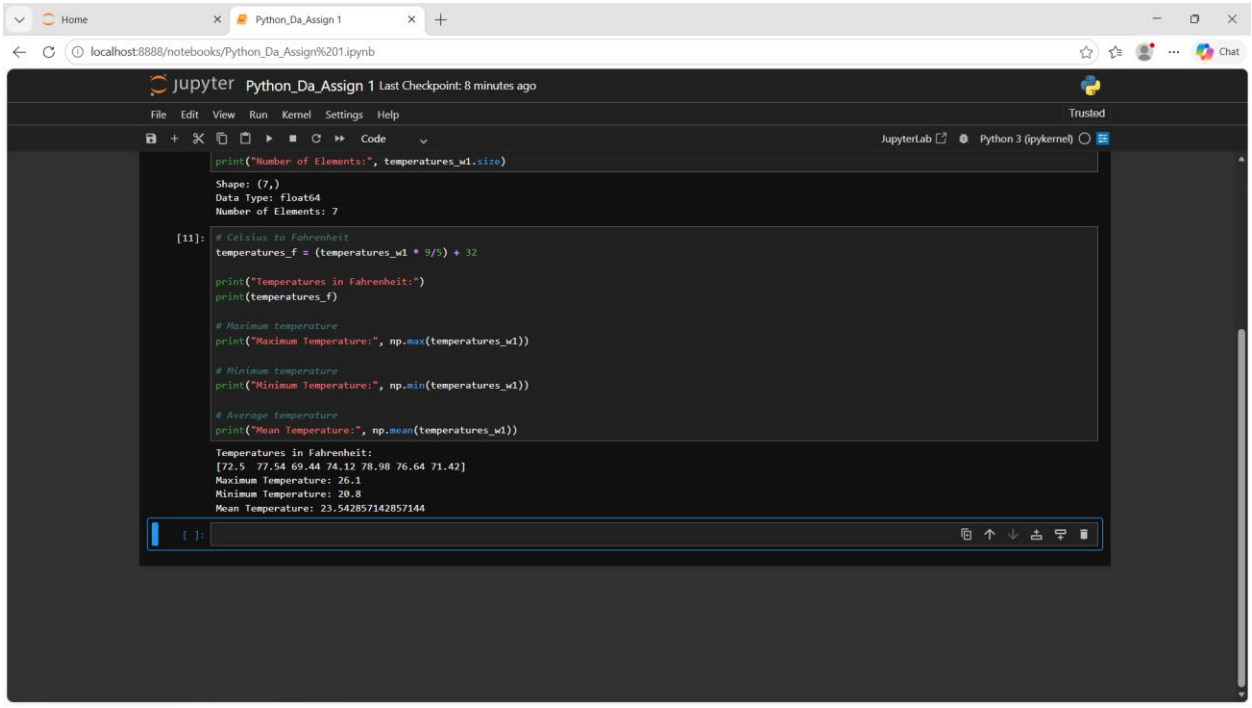
1. NumPy Array Operations



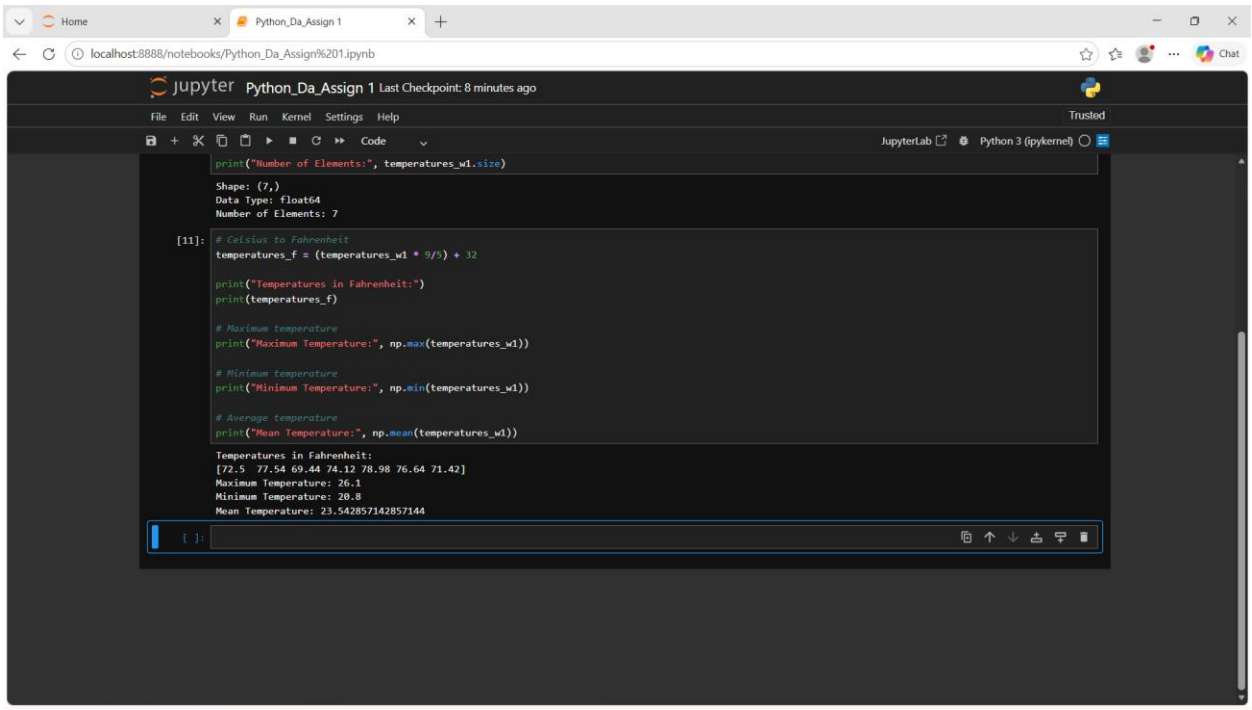
2. Exploring Array Attributes



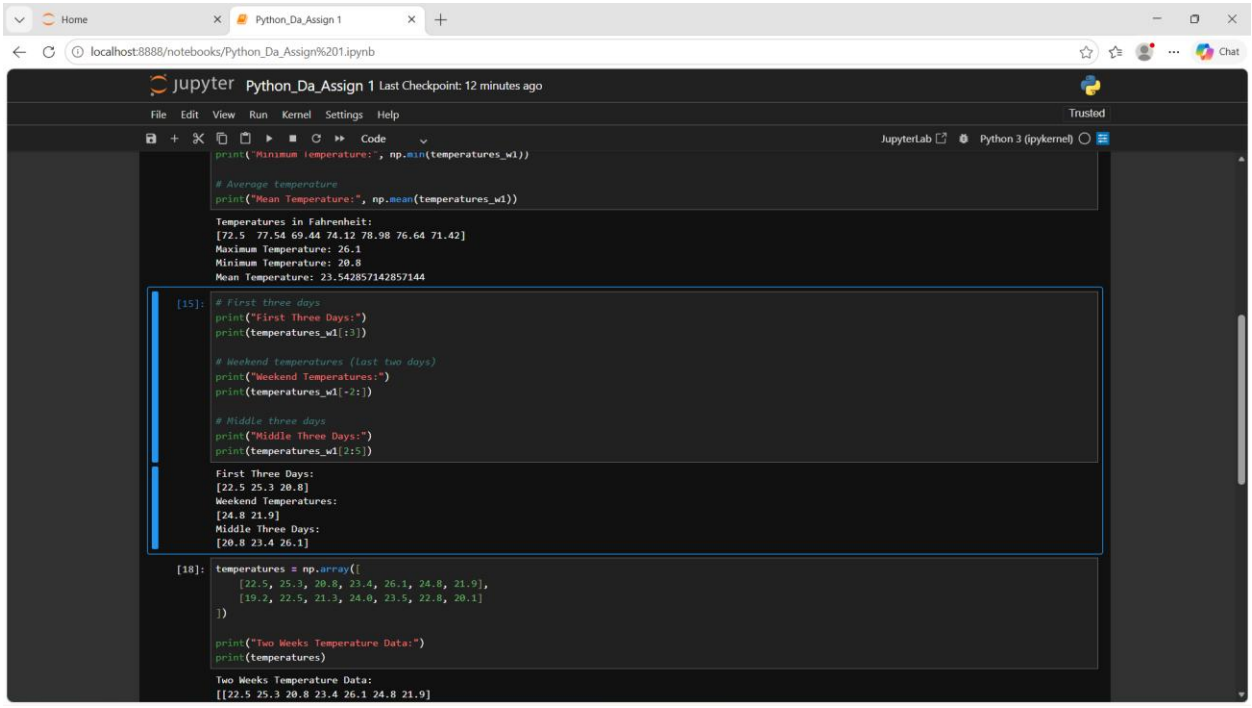
3. Celsius to Fahrenheit Conversion



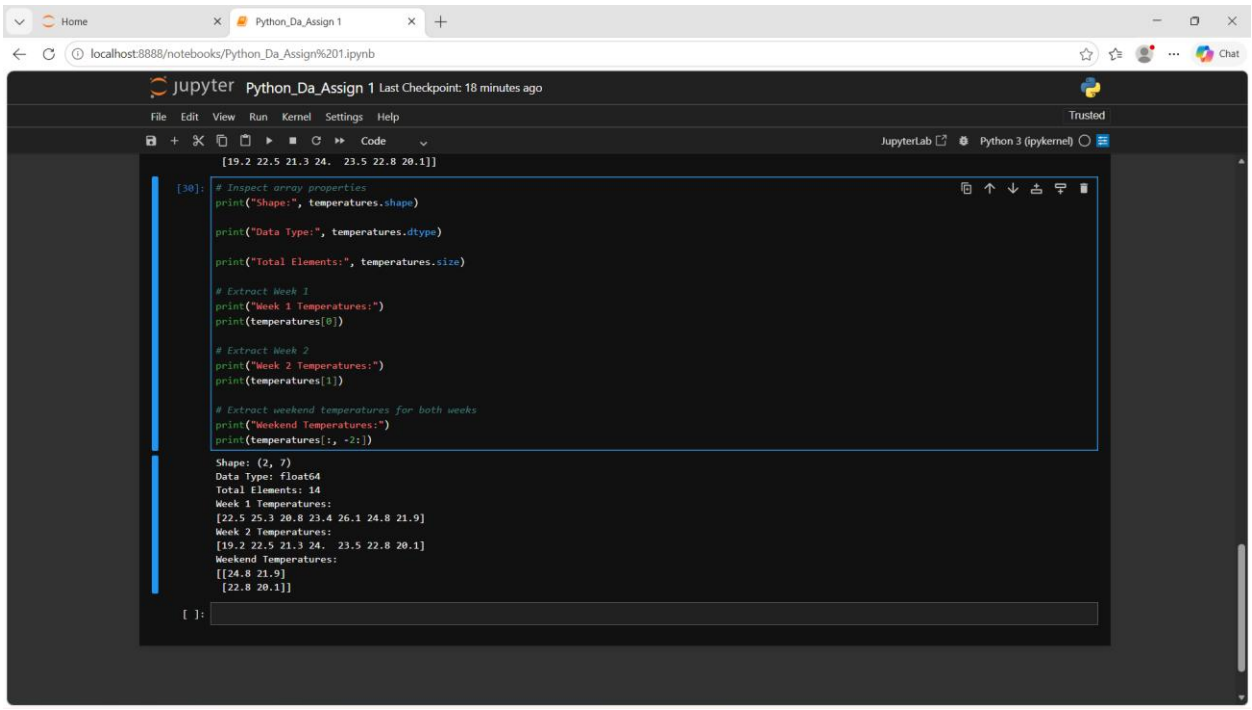
4. Statistical Operations on Temperature Data



5. Array Slicing Operations



6. Creating and Slicing a Two-Dimensional Array



Task 2: Pandas Series Operations

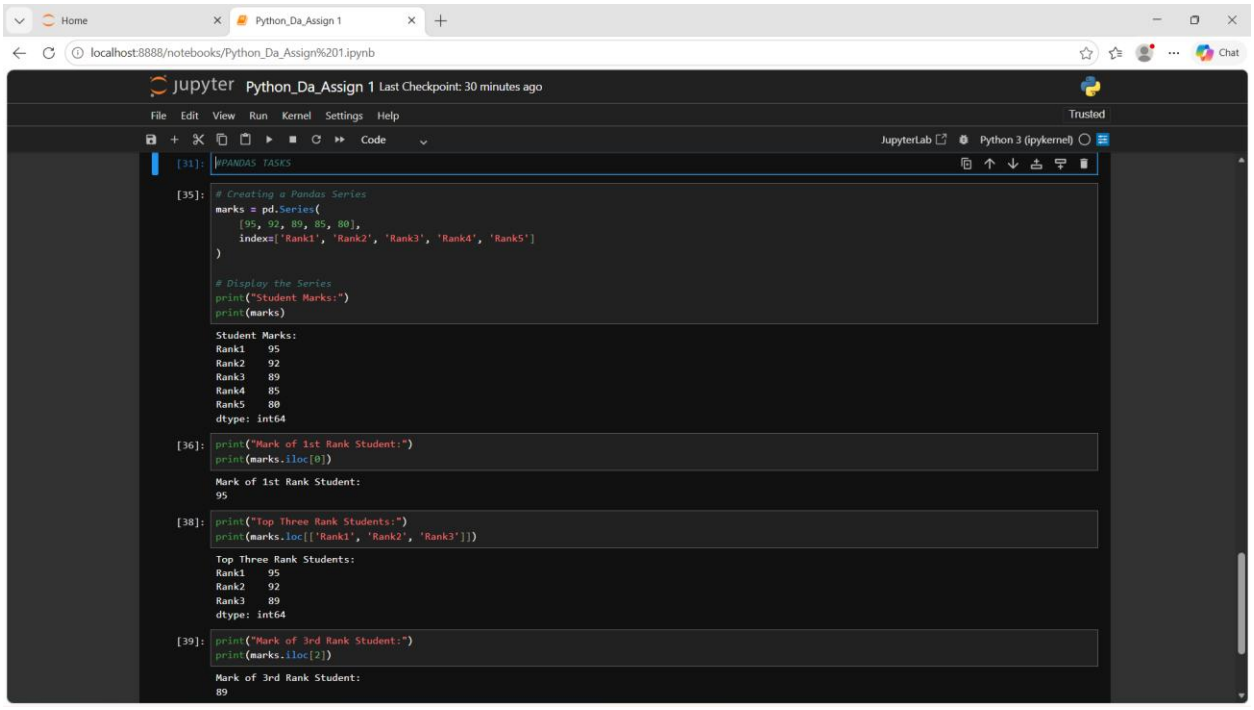
2.1 Creating a Pandas Series

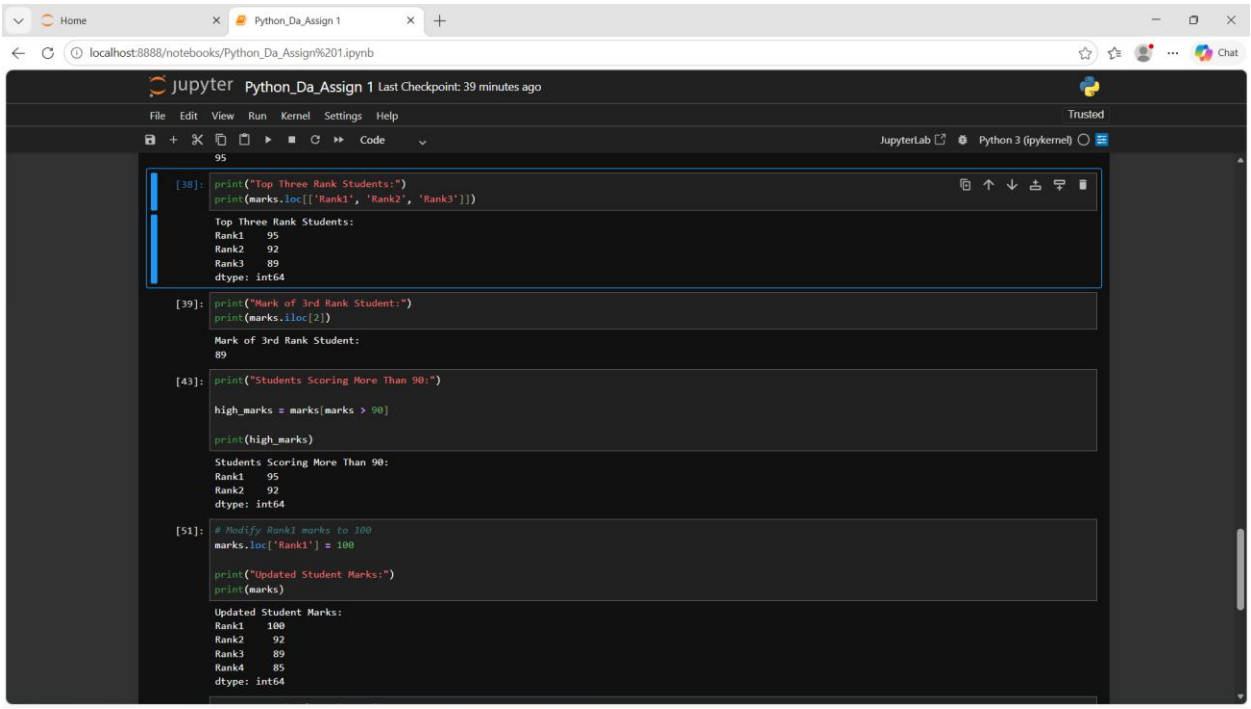
2.2 Accessing Series Elements

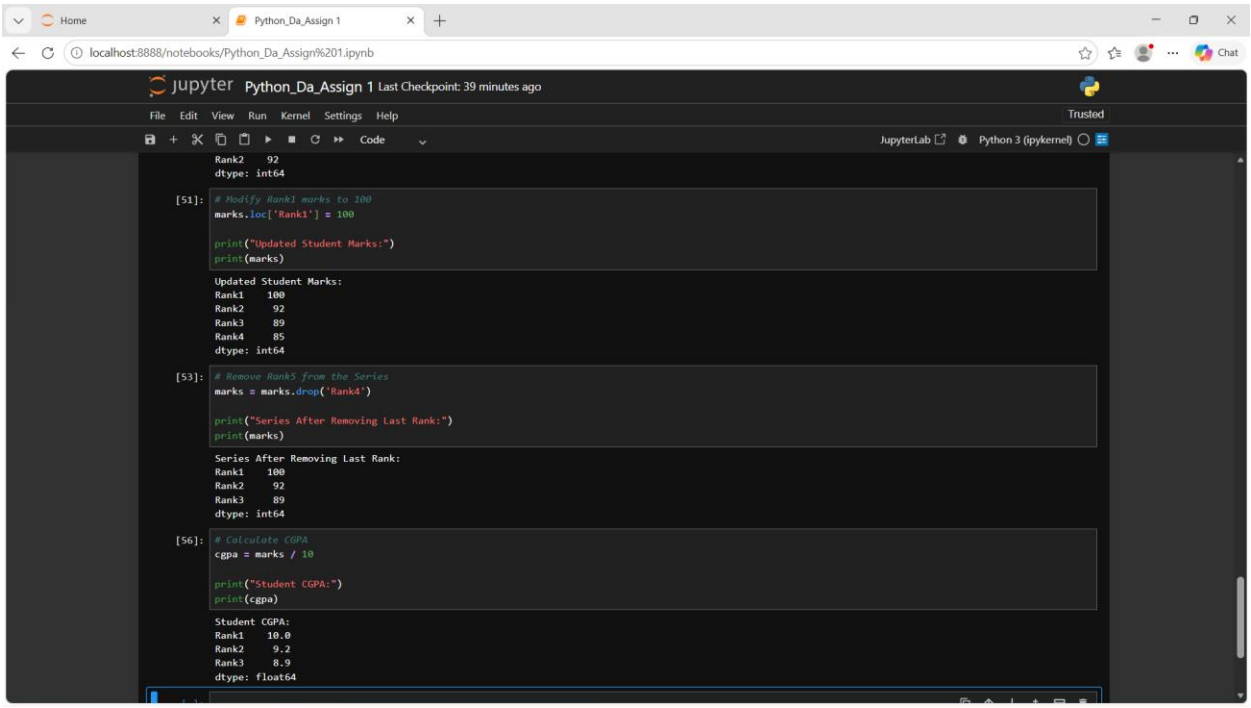
2.3 Filtering Series Using Conditions

2.4 Updating and Removing Series Elements

2.5 Calculating CGPA




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Code
JupyterLab
Python 3 (ipykernel)
95
[38]:
print("Top Three Rank Students:")
print(marks.loc[['Rank1', 'Rank2', 'Rank3']])
Top Three Rank Students:
Rank1 95
Rank2 92
Rank3 89
dtype: int64
[39]:
print("Mark of 3rd Rank Student:")
print(marks.iloc[2])
Mark of 3rd Rank Student:
89
[43]:
print("Students Scoring More Than 90:")
high_marks = marks[marks > 90]
print(high_marks)
Students Scoring More Than 90:
Rank1 95
Rank2 92
dtype: int64
[51]:
Modify Rank1 marks to 100
marks.loc['Rank1'] = 100
print("Updated Student Marks:")
print(marks)
Updated Student Marks:
Rank1 100
Rank2 92
Rank3 89
Rank4 85
dtype: int64


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Code
JupyterLab
Python 3 (ipykernel)
Rank2 92
dtype: int64
[51]:
Modify Rank1 marks to 100
marks.loc['Rank1'] = 100
print("Updated Student Marks:")
print(marks)
Updated Student Marks:
Rank1 100
Rank2 92
Rank3 89
Rank4 85
dtype: int64
[53]:
Remove Rank5 from the Series
marks = marks.drop('Rank4')
print("Series After Removing Last Rank:")
print(marks)
Series After Removing Last Rank:
Rank1 100
Rank2 92
Rank3 89
dtype: int64
[56]:
Calculate CGPA
cgpa = marks / 10
print("Student CGPA:")
print(cgpa)
Student CGPA:
Rank1 10.0
Rank2 9.2
Rank3 8.9
dtype: float64

Task3: Pandas DataFrame Operations

3.1 Creating a Transaction DataFrame

3.2 Exploring DataFrame Structure

3.3 Selecting DataFrame Columns

3.4 Selecting Rows and Columns Using loc and iloc

3.5 Filtering Transaction Records

3.6 Analysing Product Categories and Regions

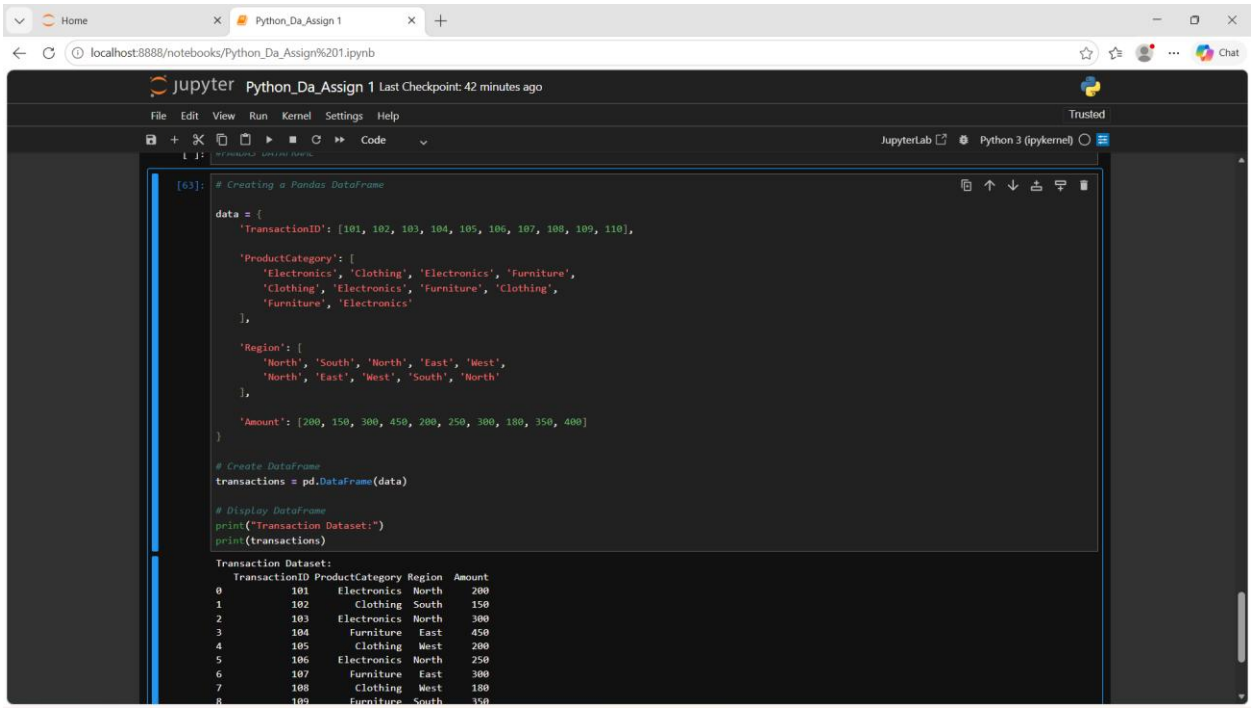
3.7 Updating Transaction Records

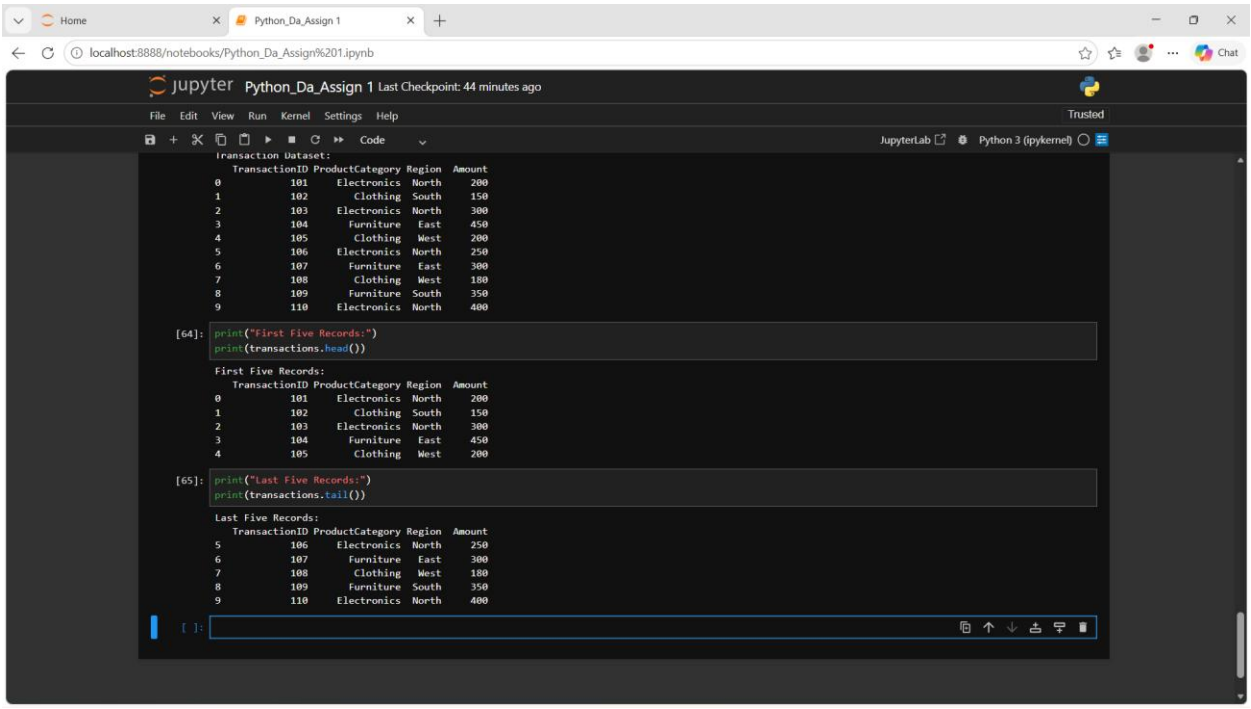
3.8 Creating a Discount Column

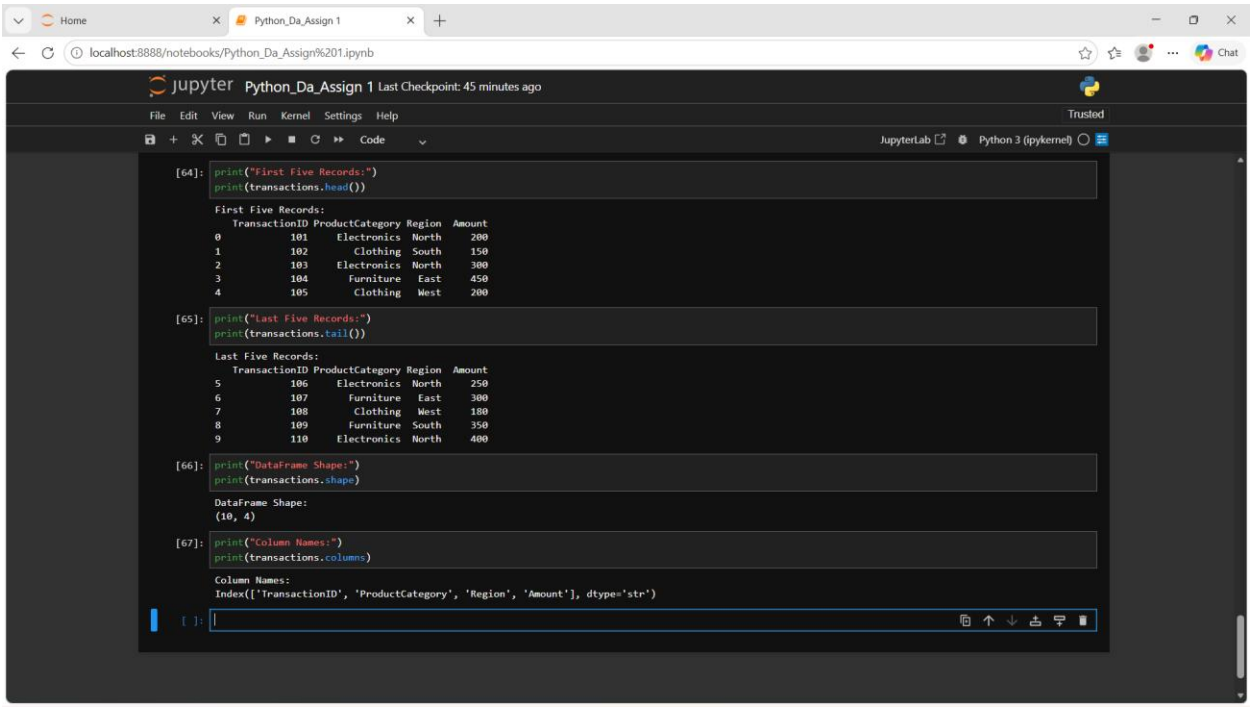
3.9 Removing a Transaction Record

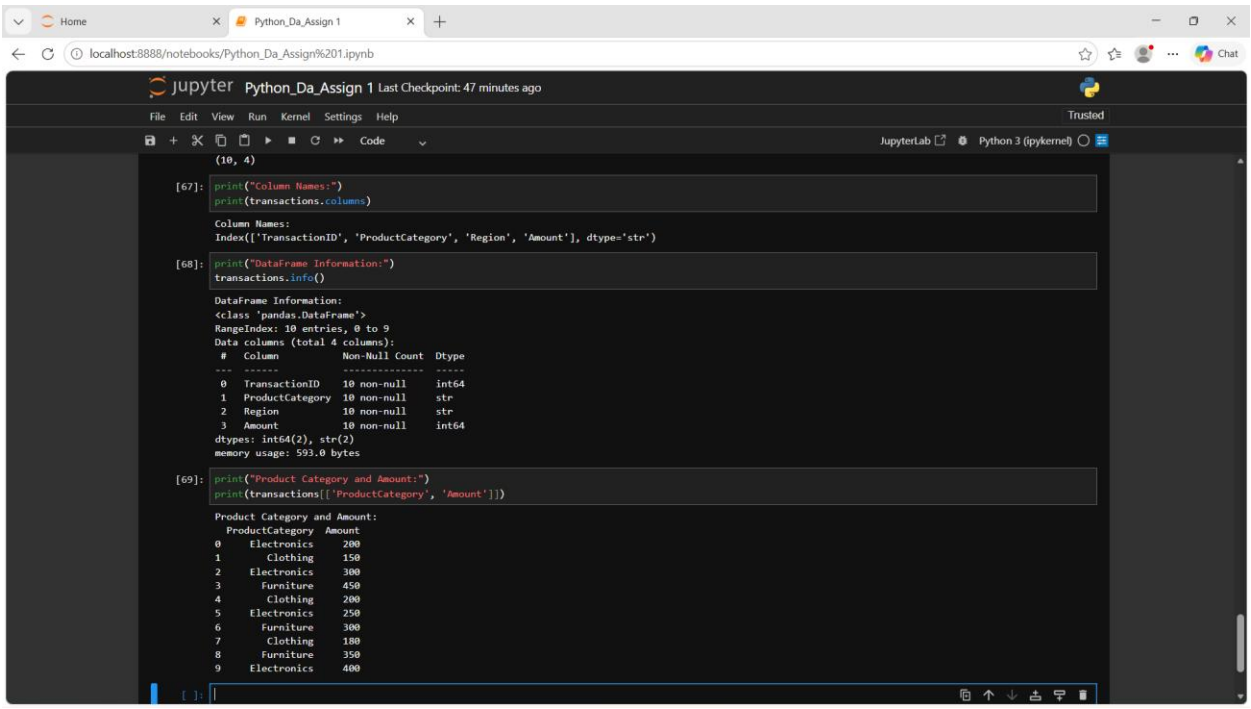
3.10 Deleting the Discount Column

3.11 Final Transaction DataFrame




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Code
JupyterLab
Python 3 (ipykernel)
Transaction Dataset:
TransactionID ProductCategory Region Amount
0 101 Electronics North 200
1 102 Clothing South 150
2 103 Electronics North 300
3 104 Furniture East 450
4 105 Clothing West 200
5 106 Electronics North 250
6 107 Furniture East 300
7 108 Clothing West 180
8 109 Furniture South 350
9 110 Electronics North 400
[64]: print("First Five Records:")
print(transactions.head())
First Five Records:
TransactionID ProductCategory Region Amount
0 101 Electronics North 200
1 102 Clothing South 150
2 103 Electronics North 300
3 104 Furniture East 450
4 105 Clothing West 200
[65]: print("Last Five Records:")
print(transactions.tail())
Last Five Records:
TransactionID ProductCategory Region Amount
5 106 Electronics North 250
6 107 Furniture East 300
7 108 Clothing West 180
8 109 Furniture South 350
9 110 Electronics North 400
[ ]:


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Code
JupyterLab
Python 3 (ipykernel)
[64]: print("First Five Records:")
print(transactions.head())
First Five Records:
TransactionID ProductCategory Region Amount
0 101 Electronics North 200
1 102 Clothing South 150
2 103 Electronics North 300
3 104 Furniture East 450
4 105 Clothing West 200
[65]: print("Last Five Records:")
print(transactions.tail())
Last Five Records:
TransactionID ProductCategory Region Amount
5 106 Electronics North 250
6 107 Furniture East 300
7 108 Clothing West 180
8 109 Furniture South 350
9 110 Electronics North 400
[66]: print("DataFrame Shape:")
print(transactions.shape)
DataFrame Shape:
(10, 4)
[67]: print("Column Names:")
print(transactions.columns)
Column Names:
Index(['TransactionID', 'ProductCategory', 'Region', 'Amount'], dtype='str')
[ ]:



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Code — JupyterLab — Python 3 (ipykernel)

```
(10, 4)
```

[67]:
```
print("Column Names:")
print(transactions.columns)
```

```
Column Names:
Index(['TransactionID', 'ProductCategory', 'Region', 'Amount'], dtype='str')
```

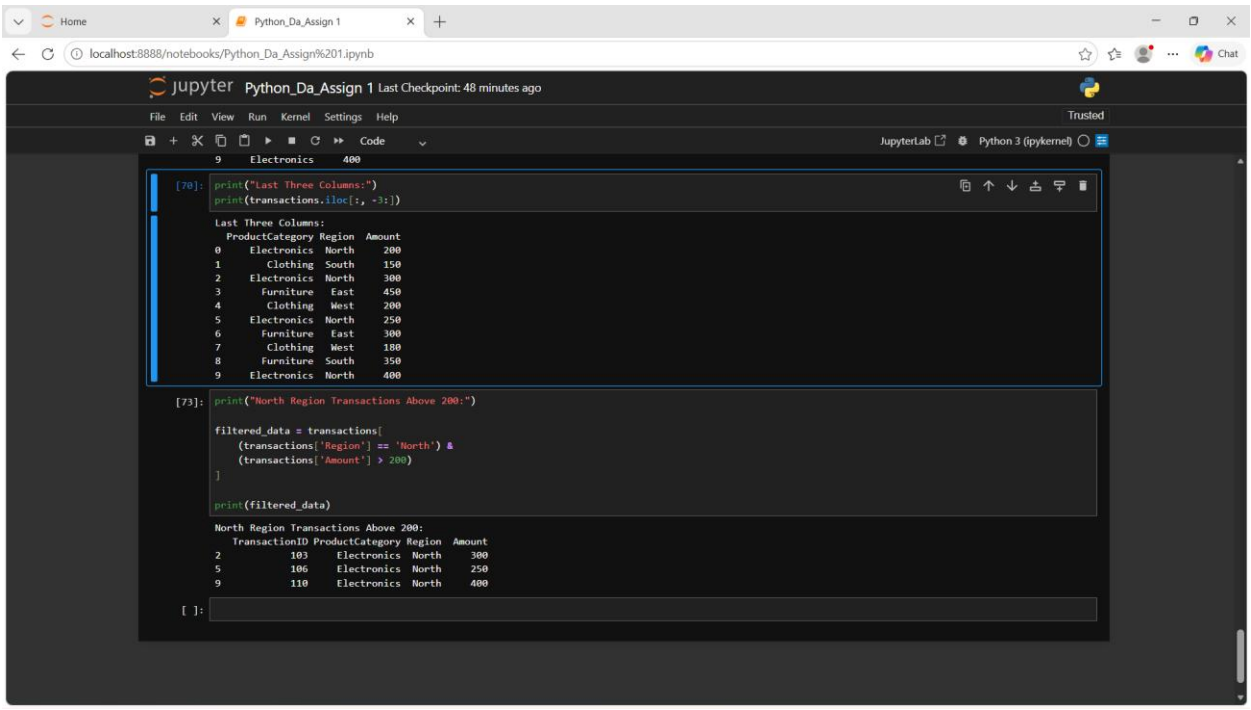
[68]:
```
print("DataFrame Information:")
transactions.info()
```

```
DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   TransactionID   10 non-null     int64
 1   ProductCategory 10 non-null     str
 2   Region          10 non-null     str
 3   Amount          10 non-null     int64
dtypes: int64(2), str(2)
memory usage: 593.0 bytes
```

[69]:
```
print("Product Category and Amount:")
print(transactions[['ProductCategory', 'Amount']])
```

```
Product Category and Amount:
  ProductCategory  Amount
0     Electronics     200
1        Clothing     150
2     Electronics     300
3       Furniture     450
4        Clothing     200
5     Electronics     250
6       Furniture     300
7        Clothing     180
8       Furniture     350
9     Electronics     400
```

[ ]:



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Code — JupyterLab — Python 3 (ipykernel)

```
9     Electronics     400
```

[70]:
```
print("Last Three Columns:")
print(transactions.iloc[:, -3:])
```

```
Last Three Columns:
  ProductCategory Region  Amount
0     Electronics  North     200
1        Clothing  South     150
2     Electronics  North     300
3       Furniture   East     450
4        Clothing   West     200
5     Electronics  North     250
6       Furniture   East     300
7        Clothing   West     180
8       Furniture  South     350
9     Electronics  North     400
```

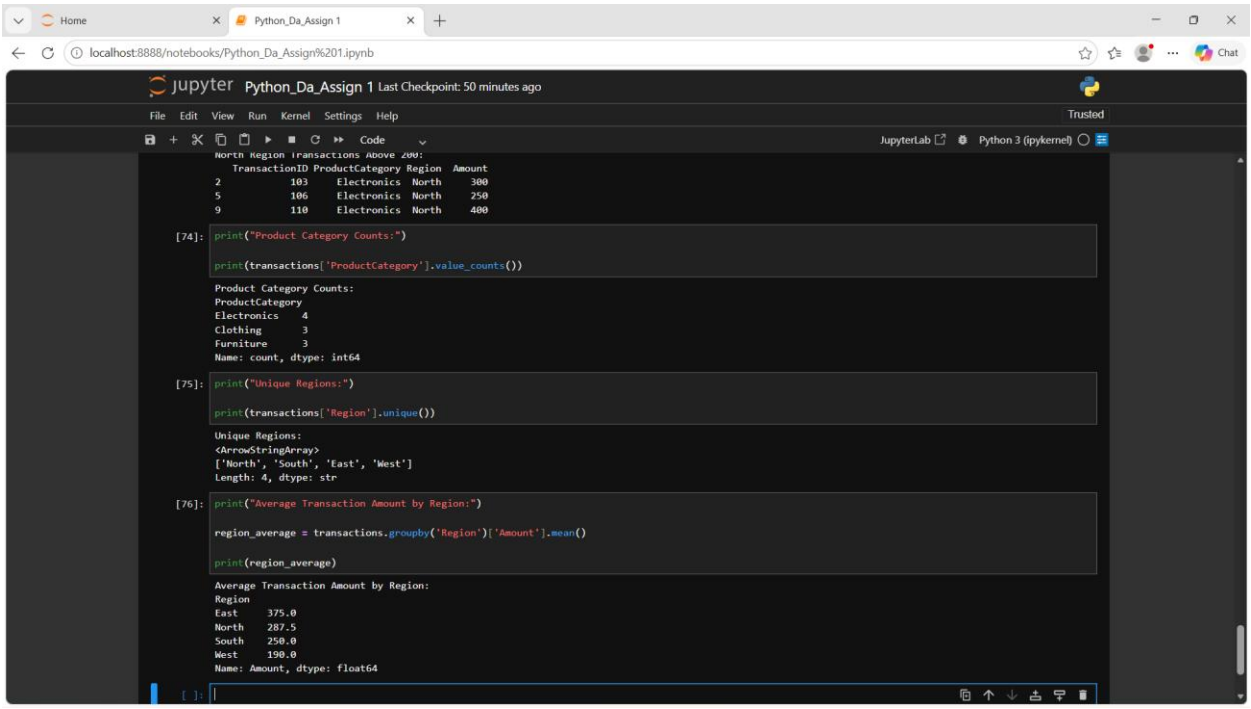
[73]:
```
print("North Region Transactions Above 200:")

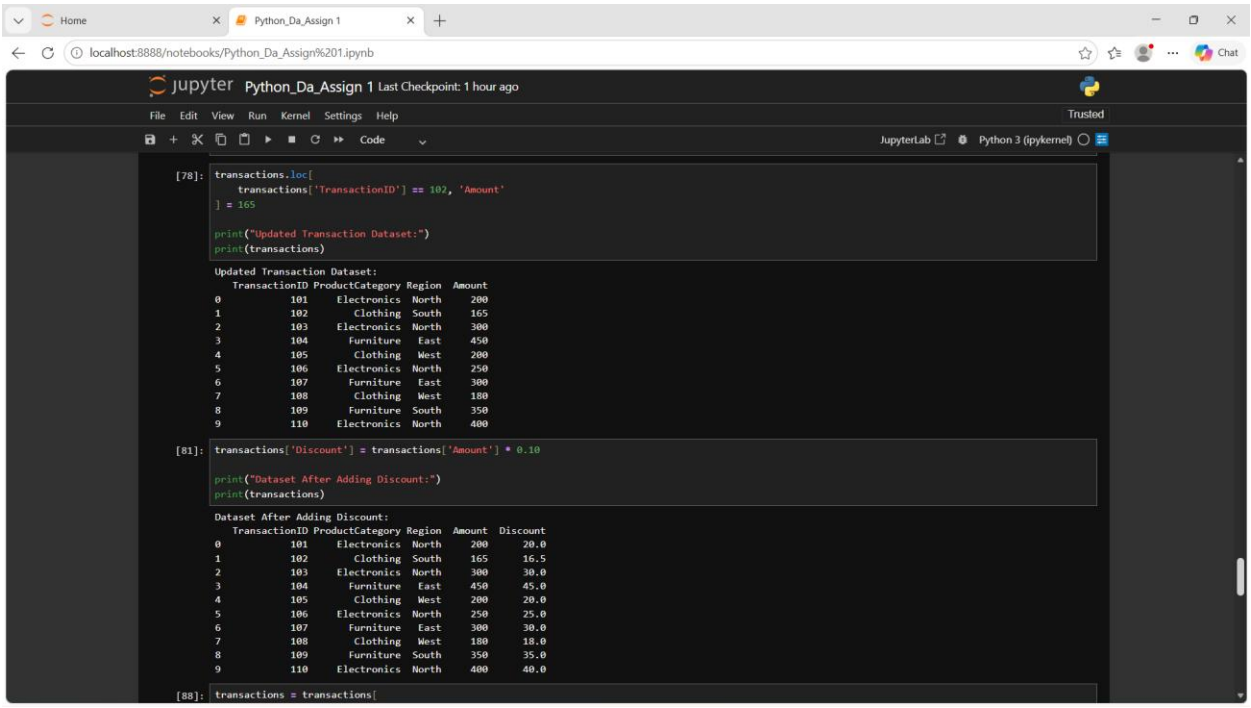
filtered_data = transactions[
    (transactions['Region'] == 'North') &
    (transactions['Amount'] > 200)
]

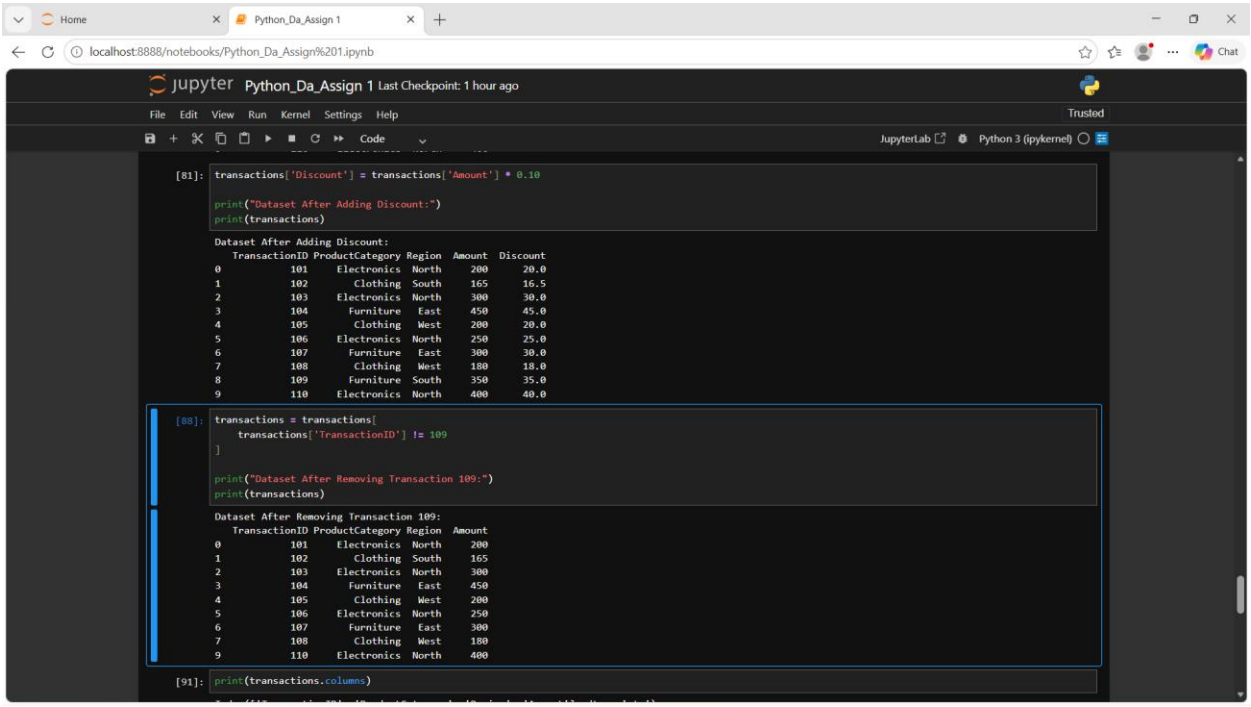
print(filtered_data)
```

```
North Region Transactions Above 200:
   TransactionID ProductCategory Region  Amount
2            103     Electronics  North     300
5            106     Electronics  North     250
9            110     Electronics  North     400
```

[ ]:


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Code
JupyterLab
Python 3 (ipykernel)
North Region Transactions Above 200:
TransactionID ProductCategory Region Amount
2 103 Electronics North 300
5 106 Electronics North 250
9 110 Electronics North 400
[74]: print("Product Category Counts:")
print(transactions['ProductCategory'].value_counts())
Product Category Counts:
ProductCategory
Electronics 4
Clothing 3
Furniture 3
Name: count, dtype: int64
[75]: print("Unique Regions:")
print(transactions['Region'].unique())
Unique Regions:
<ArrowStringArray>
['North', 'South', 'East', 'West']
Length: 4, dtype: str
[76]: print("Average Transaction Amount by Region:")
region_average = transactions.groupby('Region')['Amount'].mean()
print(region_average)
Average Transaction Amount by Region:
Region
East 375.0
North 287.5
South 250.0
West 190.0
Name: Amount, dtype: float64
[ ]:


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Code
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Python 3 (ipykernel)
[78]: transactions.loc[
transactions['TransactionID'] == 102, 'Amount'
] = 165
print("Updated Transaction Dataset:")
print(transactions)
Updated Transaction Dataset:
TransactionID ProductCategory Region Amount
0 101 Electronics North 200
1 102 Clothing South 165
2 103 Electronics North 300
3 104 Furniture East 450
4 105 Clothing West 200
5 106 Electronics North 250
6 107 Furniture East 300
7 108 Clothing West 180
8 109 Furniture South 350
9 110 Electronics North 400
[81]: transactions['Discount'] = transactions['Amount'] * 0.10
print("Dataset After Adding Discount:")
print(transactions)
Dataset After Adding Discount:
TransactionID ProductCategory Region Amount Discount
0 101 Electronics North 200 20.0
1 102 Clothing South 165 16.5
2 103 Electronics North 300 30.0
3 104 Furniture East 450 45.0
4 105 Clothing West 200 20.0
5 106 Electronics North 250 25.0
6 107 Furniture East 300 30.0
7 108 Clothing West 180 18.0
8 109 Furniture South 350 35.0
9 110 Electronics North 400 40.0
[88]: transactions = transactions[


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Code
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Python 3 (ipykernel)
[81]:
transactions['Discount'] = transactions['Amount'] * 0.10
print("Dataset After Adding Discount:")
print(transactions)
Dataset After Adding Discount:
TransactionID ProductCategory Region Amount Discount
0 101 Electronics North 200 20.0
1 102 Clothing South 165 16.5
2 103 Electronics North 300 30.0
3 104 Furniture East 450 45.0
4 105 Clothing West 200 20.0
5 106 Electronics North 250 25.0
6 107 Furniture East 300 30.0
7 108 Clothing West 180 18.0
8 109 Furniture South 350 35.0
9 110 Electronics North 400 40.0
[88]:
transactions = transactions[
transactions['TransactionID'] != 109
]
print("Dataset After Removing Transaction 109:")
print(transactions)
Dataset After Removing Transaction 109:
TransactionID ProductCategory Region Amount
0 101 Electronics North 200
1 102 Clothing South 165
2 103 Electronics North 300
3 104 Furniture East 450
4 105 Clothing West 200
5 106 Electronics North 250
6 107 Furniture East 300
7 108 Clothing West 180
9 110 Electronics North 400
[91]:
print(transactions.columns)